MYSQL PROJECT

TOPIC: AIR QUALITY ANALYSIS OF INDIAN CITIES (2015–2020) USING MYSQL

SOURCE: KAGGLE

INTRODUCTION

This project performs a structured analysis of daily air quality data from Indian cities using MySQL. The dataset, published by the Central Pollution Control Board (CPCB), contains 29,531 daily records for 26 cities from January 2015 to July 2020. It includes pollutant levels (PM2.5, PM10, NO2, SO2, CO, O3 and others), the Air Quality Index (AQI) and the AQI category.

The objective of this analysis is to extract meaningful insights on pollution levels across cities, seasonal and yearly trends, and the effect of the COVID-19 lockdown, using SQL techniques such as data cleaning, filtering, aggregation, subqueries, CTEs and window functions.

TOOL: MySQL Workbench

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Data Preparation and Cleaning

The CSV file was imported into MySQL Workbench as text, so the data was prepared first. Empty cells were converted to NULL, pollutant columns were converted to numbers, and a proper DATE column was added.

```
UPDATE city_day SET PM10 = NULL WHERE PM10 = '';  -- repeated for every pollutant column, AQI and AQI_Bucket

ALTER TABLE city_day MODIFY PM2_5 DOUBLE, MODIFY PM10 DOUBLE, MODIFY NO_ DOUBLE, MODIFY NO2 DOUBLE, MODIFY NOx DOUBLE, MODIFY NH3 DOUBLE, MODIFY CO DOUBLE, MODIFY SO2 DOUBLE, MODIFY O3 DOUBLE, MODIFY Benzene DOUBLE, MODIFY Toluene DOUBLE, MODIFY Xylene DOUBLE, MODIFY AQI DOUBLE;

ALTER TABLE city_day ADD COLUMN Date_Clean DATE; UPDATE city_day SET Date_Clean = STR_TO_DATE(Date, '%Y-%m-%d');
```

A. Count the missing values in each column

```
SELECT COUNT(*) AS total_rows, COUNT(*)-COUNT(PM2_5) AS pm25_nulls, COUNT(*)-COUNT(PM10) AS pm10_nulls, COUNT(*)-COUNT(NO2) AS no2_nulls, COUNT(*)-COUNT(NH3) AS nh3_nulls, COUNT(*)-COUNT(Xylene) AS xylene_nulls, COUNT(*)-COUNT(AQI) AS aqi_nulls FROM city_day;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_rows	pm25_nulls	pm10_nulls	no2_nulls	nh3_nulls	xylene_nulls	aqi_nulls
29531	4598	11140	3585	10328	19856	4681

Insight: Missing values are common in the CPCB data. Xylene (18,109 rows, 61%), PM10 (11,140 rows, 38%) and NH3 (10,328 rows, 35%) have the most gaps, and AQI itself is missing on 4,681 rows (16%). Blanks were stored as NULL so that averages ignore them instead of treating them as zero.

B. Check for duplicate city and date records

```
SELECT City, Date_Clean, COUNT(*) FROM city_day GROUP BY City, Date_Clean HAVING COUNT(*) > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	Date_Clean	COUNT(*)

Insight: The query returned no rows, so there are no duplicate city-date records. Each city has at most one reading per day.

Queries

1. **Write a query to select the first 10 records showing City, Date, PM2.5, PM10, AQI and AQI Bucket**

   SELECT City, Date_Clean, PM2_5, PM10, AQI, AQI_Bucket FROM city_day LIMIT 10;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	Date_Clean	PM2_5	PM10	AQI	AQI_Bucket
Ahmedabad	2015-01-01	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-02	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-03	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-04	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-05	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-06	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-07	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-08	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-09	NULL	NULL	NULL	NULL
Ahmedabad	2015-01-10	NULL	NULL	NULL	NULL

Insight: The first records are Ahmedabad readings from January 2015. PM2.5, PM10 and AQI are NULL for these days, which shows that early records have missing pollutant data and why the cleaning step was needed.

2. **Retrieve all records for the city of Delhi**

   SELECT * FROM city_day WHERE City = 'Delhi' ORDER BY Date_Clean;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

City	Date	PM2_5	PM10	NO_	NO2	NOx	NH3	CO	SO2	O3	Benzen
Delhi	2016-01-29	311.08	498.38	140.59	117.07	139.36	74.39	2.54	29.99	175.04	17.15
Delhi	2015-01-01	313.22	607.98	69.16	36.39	110.59	33.85	15.2	9.25	41.68	14.36
Delhi	2016-02-05	206.72	383.25	103.55	86.15	180.54	38.98	1.82	30.2	121.22	13.75
Delhi	2016-01-24	303.5	446.28	18.81	64.55	84.23	56.64	2.6	14.74	132.54	13.06
Delhi	2016-01-28	283.03	463.03	96.44	90.45	154.28	68.84	2.09	23.63	151.6	13.06
Delhi	2016-10-29	350.39	516.59	208.23	162.5	167.24	79.24	2.65	54.38	139.22	13.02
Delhi	2016-04-09	166.76	272.67	65.1	63.03	122.4	35.12	1.51	46.97	47.93	12.58
Delhi	2016-01-30	285.83	450.72	114.59	111.19	202.98	78.01	2.54	27.94	157.74	11.99
Delhi	2016-01-23	294.3	480	58.3	78.21	130.91	67.05	2.13	16.81	147.56	11.67
Delhi	2016-11-03	482.4	483.99	127.32	148.92	116.45	104.84	2.78	29.4	108.33	11.64
Delhi	2016-02-06	224.94	348.14	139.49	85.86	143.3	41.58	2.6	26.22	149.67	11.57
Delhi	2016-11-02	537.96	693.21	121.1	149.12	157.33	111.69	2.8	26.66	160.01	11.45
Delhi	2015-10-05	180.52	347.95	80.34	75.42	116.95	36.94	3.15	17.32	45.8	10.88
Delhi	2016-03-03	165.22	264.31	116.23	100.71	182.37	48.37	2.61	33.45	113.89	10.66
Delhi	2016-10-28	318.57	503	137.36	119.42	134.89	64.34	1.91	27.89	107.12	10.61
Delhi	2016-02-19	207.1	307.38	54.8	66.45	119.87	67.61	1.76	32.25	108.89	10.6

Insight: Delhi has 2,009 daily records from January 2015 to July 2020, a complete daily series. This makes it one of the most reliable cities for trend analysis.

3. Generate a list of all unique cities and AQI categories in the dataset

SELECT DISTINCT City FROM city_day ORDER BY City;

SELECT DISTINCT AQI_Bucket FROM city_day WHERE AQI_Bucket IS NOT NULL;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City
Ahmedabad
Aizawl
Amaravati
Amritsar
Bengaluru
Bhopal
Brajrajnagar
Chandigarh
Chennai
Coimbatore
Delhi
Ernakulam
Gurugram
Guwahati
Hyderabad
Jaipur
Jorapokhar
Kochi
Kolkata
Lucknow
Mumbai
Patna
Shillong

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AQI_Bucket
Poor
Very Poor
Severe
Moderate
Satisfactory
Good

Insight: The dataset covers 26 cities across India, including the Kerala cities Kochi, Ernakulam and Thiruvananthapuram. AQI is grouped into six categories: Good, Satisfactory, Moderate, Poor, Very Poor and Severe.

4. Retrieve all days with a Severe AQI category

```
SELECT City, Date_Clean, AQI FROM city_day WHERE AQI_Bucket = 'Severe' ORDER BY AQI DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

City	Date_Clean	AQI
Ahmedabad	2018-02-19	2049
Ahmedabad	2018-02-10	1917
Ahmedabad	2016-07-14	1842
Ahmedabad	2017-11-14	1747
Ahmedabad	2019-01-03	1719
Ahmedabad	2018-11-18	1672
Ahmedabad	2018-11-21	1646
Ahmedabad	2018-11-08	1630
Ahmedabad	2018-02-01	1613
Ahmedabad	2019-10-13	1595
Ahmedabad	2017-11-13	1577
Ahmedabad	2018-10-23	1561
Ahmedabad	2018-12-22	1558
Ahmedabad	2018-11-13	1548
Ahmedabad	2019-01-21	1537
Ahmedabad	2018-02-23	1526
Ahmedabad	2018-02-22	1483
Ahmedabad	2018-02-18	1477
Ahmedabad	2019-01-01	1474
Ahmedabad	2018-11-19	1472
Ahmedabad	2019-03-28	1459
Ahmedabad	2018-03-31	1434
Ahmedabad	2017-11-28	1430

Insight: There are 1,338 Severe days in total, with Ahmedabad (638), Delhi (239) and Patna (174) accounting for most of them. The highest value is an AQI of 2,049 in Ahmedabad on 19 February 2018, which is far above the normal 0–500 scale and is likely a data anomaly.

5. Calculate the overall average PM2.5, PM10 and AQI

```
SELECT ROUND(AVG(PM2_5),1) AS avg_pm25, ROUND(AVG(PM10),1) AS avg_pm10, ROUND(AVG(AQI),1) AS avg_aqi FROM city_day;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_pm25	avg_pm10	avg_aqi
67.5	118.1	166.5

Insight: The average AQI is 166.5, which falls in the Moderate category. Average PM2.5 is 67.5 and PM10 is 118.1, which is well above the WHO 24-hour guideline of 15 µg/m³ for PM2.5.

6. Count the number of days for each combination of City and AQI Bucket, ordered by highest count

SELECT City, AQI_Bucket, COUNT(*) AS days FROM city_day WHERE AQI_Bucket IS NOT NULL GROUP BY City, AQI_Bucket ORDER BY days DESC;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	AQI_Bucket	days
Bengaluru	Satisfactory	1124
Hyderabad	Moderate	958
Chennai	Satisfactory	941
Chennai	Moderate	804
Thiruvananthapuram	Satisfactory	768
Hyderabad	Satisfactory	710
Jaipur	Moderate	651
Ahmedabad	Severe	638
Bengaluru	Moderate	630
Visakhapatnam	Moderate	580
Lucknow	Moderate	578
Delhi	Poor	542
Delhi	Very Poor	520
Delhi	Moderate	519
Patna	Moderate	507
Amritsar	Satisfactory	473
Lucknow	Very Poor	473
Gurugram	Moderate	454
Jorapokhar	Moderate	451
Amritsar	Moderate	448
Visakhapatnam	Satisfactory	447

Insight: The most common combinations are Bengaluru (1,124 Satisfactory days), Hyderabad (958 Moderate days) and Chennai (941 Satisfactory days). Southern cities spend most of their days in the Satisfactory and Moderate categories.

7. Find the top 5 most polluted cities by average AQI

SELECT City, ROUND(AVG(AQI),1) AS avg_aqi FROM city_day WHERE AQI IS NOT NULL GROUP BY City ORDER BY avg_aqi DESC LIMIT 5;

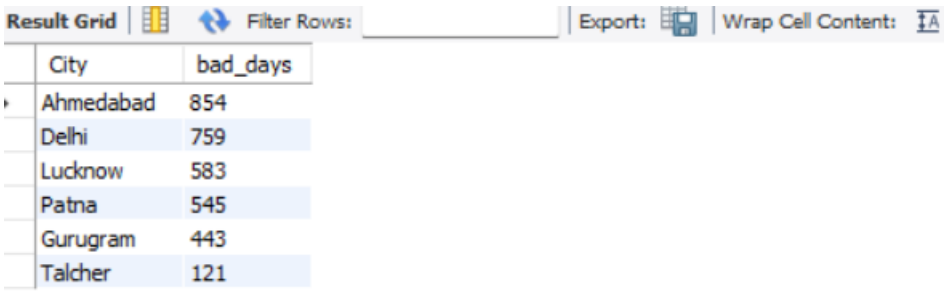
Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

City	avg_aqi
Ahmedabad	452.1
Delhi	259.5
Patna	240.8
Gurugram	225.1
Lucknow	218

Insight: Ahmedabad (452.1), Delhi (259.5), Patna (240.8), Gurugram (225.1) and Lucknow (218.0) have the highest average AQI. Ahmedabad's average is inflated by extreme readings between 2017 and 2019, so Delhi, Patna, Gurugram and Lucknow are the more reliable picture of the northern pollution belt.

8. Identify cities with more than 100 Severe or Very Poor days

```
SELECT City, COUNT(*) AS bad_days FROM city_day WHERE AQI_Bucket IN
('Severe','Very Poor') GROUP BY City HAVING COUNT(*) > 100 ORDER BY
bad_days DESC;
```

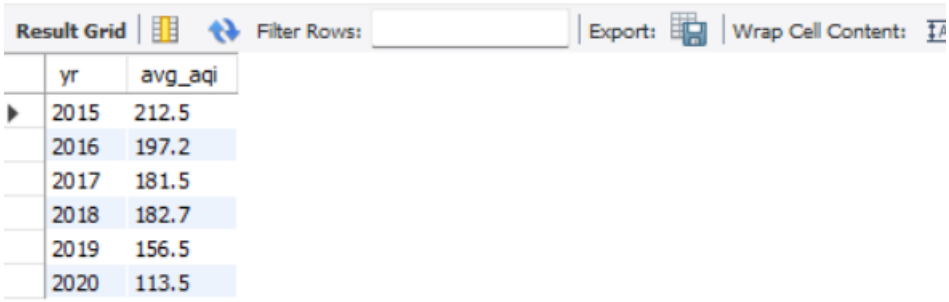


City	bad_days
Ahmedabad	854
Delhi	759
Lucknow	583
Patna	545
Gurugram	443
Talcher	121

Insight: Six cities crossed 100 Severe or Very Poor days: Ahmedabad (854), Delhi (759), Lucknow (583), Patna (545), Gurugram (443) and Talcher (121). Most of them are in north India.

9. Calculate the average AQI for each year

```
SELECT YEAR(Date_Clean) AS yr, ROUND(AVG(AQI),1) AS avg_aqi FROM city_day
WHERE AQI IS NOT NULL GROUP BY yr ORDER BY yr;
```

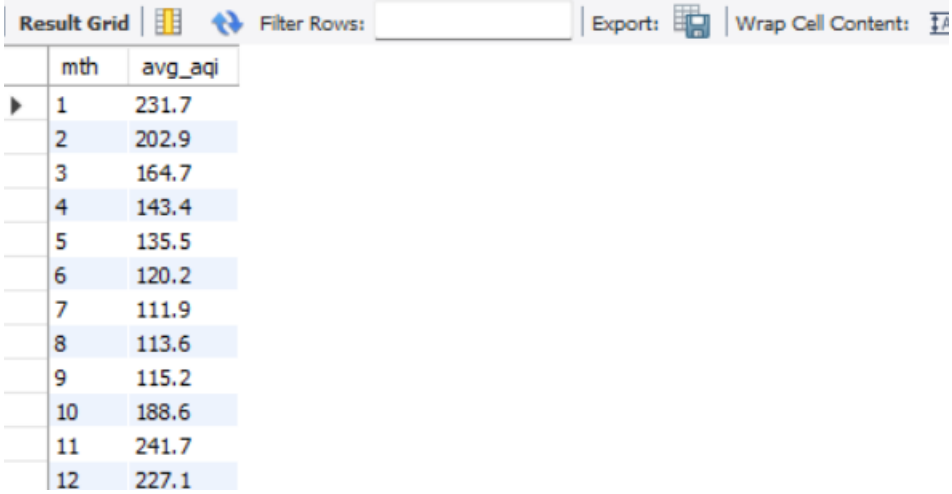


yr	avg_aqi
2015	212.5
2016	197.2
2017	181.5
2018	182.7
2019	156.5
2020	113.5

Insight: Average AQI fell from 212.5 in 2015 to 156.5 in 2019, a drop of about 26%. The 2020 value of 113.5 is lower still, but 2020 covers only January to July and includes the lockdown, so it is not comparable with a full year.

10. Calculate the average AQI for each month to find the seasonal pattern

```
SELECT MONTH(Date_Clean) AS mth, ROUND(AVG(AQI),1) AS avg_aqi FROM city_day WHERE AQI IS NOT NULL GROUP BY mth ORDER BY mth;
```



mth	avg_aqi
1	231.7
2	202.9
3	164.7
4	143.4
5	135.5
6	120.2
7	111.9
8	113.6
9	115.2
10	188.6
11	241.7
12	227.1

Insight: Pollution is strongly seasonal. AQI peaks in November (241.7), December (227.1) and January (231.7), and is lowest in July (111.9). Winter pollution is almost double the monsoon level.

11. Find cities whose average AQI is above the overall average AQI

```
SELECT City, ROUND(AVG(AQI),1) AS avg_aqi FROM city_day WHERE AQI IS NOT NULL GROUP BY City HAVING AVG(AQI) > (SELECT AVG(AQI) FROM city_day) ORDER BY avg_aqi DESC;
```



City	avg_aqi
Ahmedabad	452.1
Delhi	259.5
Patna	240.8
Gurugram	225.1
Lucknow	218
Talcher	172.9

Insight: Only 6 of the 26 cities are above the overall average of 166.5: Ahmedabad, Delhi, Patna, Gurugram, Lucknow and Talcher (172.9). The remaining 20 cities are below it.

12. Calculate the percentage of days in each AQI category

```
SELECT AQI_Bucket, COUNT(*) AS days, ROUND(COUNT(*)*100 / (SELECT COUNT(*) FROM city_day WHERE AQI_Bucket IS NOT NULL),2) AS pct FROM city_day WHERE AQI_Bucket IS NOT NULL GROUP BY AQI_Bucket ORDER BY days DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

AQI_Bucket	days	pct
Moderate	8829	35.53
Satisfactory	8224	33.09
Poor	2781	11.19
Very Poor	2337	9.40
Good	1341	5.40
Severe	1338	5.38

Insight: Moderate (35.5%) and Satisfactory (33.1%) are the most common categories. Only 5.4% of days were Good, while 26% were Poor, Very Poor or Severe, so roughly one day in four had unhealthy air.

13. Find the top 3 cities by average PM2.5

```
WITH city_pm AS (SELECT City, AVG(PM2_5) AS avg_pm25 FROM city_day WHERE PM2_5 IS NOT NULL GROUP BY City) SELECT City, ROUND(avg_pm25,1) AS avg_pm25 FROM city_pm ORDER BY avg_pm25 DESC LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	avg_pm25
Patna	123.5
Delhi	117.2
Gurugram	117.1

Insight: Patna (123.5), Delhi (117.2) and Gurugram (117.1) have the highest fine-particle (PM2.5) levels. Ahmedabad does not appear here, which supports the view that its AQI average is inflated by extreme values.

14. Find the days when PM2.5 was more than twice the overall average

```
SELECT City, Date_Clean, PM2_5 FROM city_day WHERE PM2_5 > 2 * (SELECT
AVG(PM2_5) FROM city_day) ORDER BY PM2_5 DESC LIMIT 20;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

City	Date_Clean	PM2_5
Gurugram	2017-08-29	949.99
Shillong	2019-09-03	917.77
Guwahati	2019-05-04	916.67
Gurugram	2016-03-10	914.94
Shillong	2019-09-02	914.64
Gurugram	2016-03-17	894.75
Amritsar	2017-05-08	868.66
Gurugram	2017-07-30	858.73
Amritsar	2017-05-10	832.8
Gurugram	2017-07-09	821.42
Gurugram	2017-07-16	812.49
Lucknow	2015-03-20	742.67
Gurugram	2017-07-14	723.16

Insight: The threshold is about 134.9 µg/m³, and 2,671 days exceeded it. Delhi (638), Lucknow (589) and Patna (539) have the most such days, and the highest reading in the dataset is about 950 µg/m³.

15. Calculate the 3-month moving average of AQI for Delhi

```
WITH m AS (SELECT DATE_FORMAT(Date_Clean,'%Y-%m') AS ym, AVG(AQI) AS
a FROM city_day WHERE City='Delhi' AND AQI IS NOT NULL GROUP BY ym)
SELECT ym, ROUND(a,1) AS avg_aqi, ROUND(AVG(a) OVER (ORDER BY ym
ROWS BETWEEN 2 PRECEDING AND CURRENT ROW),1) AS moving_avg_3m
FROM m;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ym	avg_aqi	moving_avg_3m
2015-01	342.3	342.3
2015-02	327.9	335.1
2015-03	256.1	308.8
2015-04	305.3	296.4
2015-05	348.6	303.3
2015-06	258.3	304.1
2015-07	200.3	269.1
2015-08	218.1	225.6
2015-09	225.3	214.6
2015-10	325	256.1
2015-11	405	318.4
2015-12	355.3	361.8
2016-01	441.9	400.8

Insight: The moving average smooths month-to-month noise and shows a clear winter peak in every year. Delhi's monthly AQI was highest in November 2016 (474.8) and lowest in July 2017 (67.2).

16. Calculate the month-over-month change in Delhi's AQI

```
WITH m AS (SELECT DATE_FORMAT(Date_Clean,'%Y-%m') AS ym, AVG(AQI) AS a FROM city_day WHERE City='Delhi' AND AQI IS NOT NULL GROUP BY ym) SELECT ym, ROUND(a,1) AS avg_aqi, ROUND(a - LAG(a) OVER (ORDER BY ym),1) AS mom_change FROM m;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ym	avg_aqi	mom_change
2015-03	256.1	-71.9
2015-04	305.3	49.2
2015-05	348.6	43.3
2015-06	258.3	-90.2
2015-07	200.3	-58
2015-08	218.1	17.8
2015-09	225.3	7.2
2015-10	325	99.7
2015-11	405	80
2015-12	355.3	-49.7
2016-01	441.9	86.6
2016-02	312.4	-129.5

Insight: The largest monthly rise was in October 2016 (+192.4), at the start of the winter pollution season. The largest fall was in June 2017 (−170.7), showing how quickly air quality improves when the weather changes.

17. Rank cities by average AQI within each year

```
WITH yearly AS (SELECT YEAR(Date_Clean) AS yr, City, AVG(AQI) AS avg_aqi FROM city_day WHERE AQI IS NOT NULL GROUP BY yr, City) SELECT yr, City, ROUND(avg_aqi,1) AS avg_aqi, RANK() OVER (PARTITION BY yr ORDER BY avg_aqi DESC) AS rnk FROM yearly ORDER BY yr, rnk;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

yr	City	avg_aqi	rnk
2015	Patna	350.6	1
2015	Ahmedabad	311	2
2015	Delhi	297	3
2015	Lucknow	202.2	4
2015	Chennai	148.3	5
2015	Hyderabad	143.4	6
2015	Bengaluru	112.6	7
2016	Ahmedabad	310.2	1
2016	Delhi	301.4	2
2016	Patna	252.3	3
2016	Lucknow	243	4
2016	Gurugram	226.7	5

Insight: Patna ranked first in 2015, and Ahmedabad ranked first from 2016 to 2020. Delhi stayed in the top three in five of the six years, which shows that its pollution problem is persistent rather than a one-year spike.

18. Compare AQI during the COVID-19 lockdown (Mar–Jun 2020) with the same months of 2019

```
WITH lockdown AS (SELECT City, AVG(CASE WHEN YEAR(Date_Clean)=2019 THEN AQI END) AS aqi_2019, AVG(CASE WHEN YEAR(Date_Clean)=2020 THEN AQI END) AS aqi_2020 FROM city_day WHERE MONTH(Date_Clean) BETWEEN 3 AND 6 AND YEAR(Date_Clean) IN (2019,2020) GROUP BY City) SELECT City, ROUND(aqi_2019,1) AS aqi_2019, ROUND(aqi_2020,1) AS aqi_2020, ROUND((aqi_2020-aqi_2019)/aqi_2019*100,1) AS pct_change FROM lockdown WHERE aqi_2019 IS NOT NULL AND aqi_2020 IS NOT NULL ORDER BY pct_change;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	aqi_2019	aqi_2020	pct_change
Ahmedabad	511.8	156.8	-69.4
Lucknow	196.9	114.3	-41.9
Delhi	212.7	131	-38.4
Amaravati	82.2	51.1	-37.9
Gurugram	195.9	128.3	-34.5
Bengaluru	105.7	72.1	-31.8
Visakhapatnam	103.6	71.9	-30.6
Hyderabad	95	67.7	-28.7
Kolkata	101.5	73.5	-27.7
Amritsar	107.8	81.5	-24.4
Patna	168.7	128	-24.2
Jaipur	129.9	98.8	-24
Mumbai	95.5	74.2	-22.3
Chennai	101.3	78.8	-22.2
Thiruvananth...	73.7	58.4	-20.8
Talcher	137	110.6	-19.3
Guwahati	125.2	104.5	-16.5
Brajrajnagar	150.2	130	-13.4
Coimbatore	66.5	61.2	-8.1

Insight: AQI fell in 19 of the 20 cities during the lockdown. Lucknow (−41.9%) and Delhi (−38.4%) improved the most among the major cities, and Bengaluru fell by 31.8%. Only Jorapokhar, an industrial town, did not improve (+2.0%). Ahmedabad's −69.4% is exaggerated by its extreme 2019 readings.

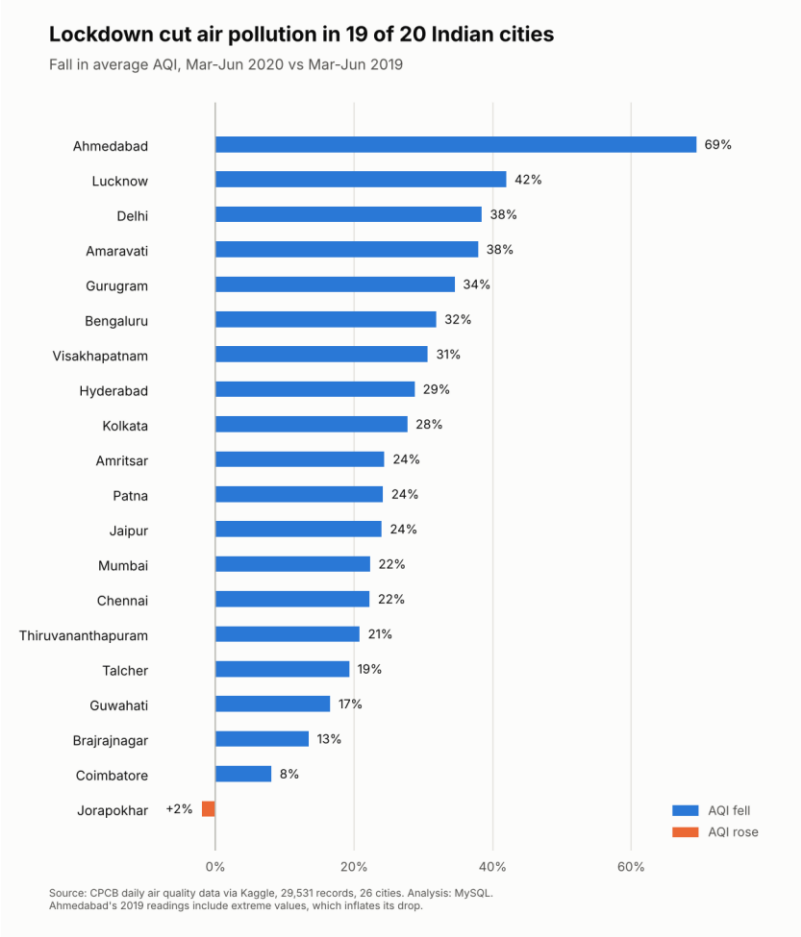


Figure: Fall in average AQI during the lockdown, by city

19. Compare air quality across the Kerala cities in the dataset

```
SELECT City, COUNT(AQI) AS days_with_aqi, ROUND(AVG(AQI),1) AS avg_aqi,
ROUND(AVG(PM2_5),1) AS avg_pm25 FROM city_day WHERE City IN
('Kochi','Ernakulam','Thiruvananthapuram') GROUP BY City ORDER BY avg_aqi
DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

City	days_with_aqi	avg_aqi	avg_pm25
Kochi	158	104.3	31.4
Ernakulam	153	92.4	24.9
Thiruvananthapuram	1052	75.9	28

Insight: Kochi has the highest average AQI (104.3), followed by Ernakulam (92.4) and Thiruvananthapuram (75.9). Kochi and Ernakulam only have data from 22 January 2020, about 155 days each, while Thiruvananthapuram has 1,052 days, so the comparison should be read with care.

CONCLUSION

The Air Quality Analysis project successfully demonstrated the use of MySQL for analysing daily air pollution data from 26 Indian cities. Through data cleaning, filtering, aggregation, GROUP BY, HAVING, subqueries, Common Table Expressions (CTEs) and window functions, the project transformed 29,531 raw CPCB records into clear, actionable insights.

The analysis showed that air quality is strongly seasonal, with AQI peaking in November to January and falling to its lowest level in the monsoon months. North Indian cities such as Delhi, Lucknow, Patna and Gurugram, along with Ahmedabad, recorded the most Severe and Very Poor days. Average AQI fell by about 26% from 2015 to 2019, and about one day in four still had Poor or worse air.

The COVID-19 lockdown provided a natural experiment: between March and June 2020, AQI fell in 19 of 20 cities compared with 2019, by 38% in Delhi and 42% in Lucknow. This shows how strongly traffic and industrial activity affect urban air quality.

The analysis has limitations. About 16% of AQI values are missing, 2020 covers only January to July, Kochi and Ernakulam have data only from January 2020, and Ahmedabad's readings between 2017 and 2019 contain extreme values. Overall, this project strengthened practical SQL skills and showed how data analysis can support environmental monitoring and policy decisions.